LFCS Additional User and Group Management Topics

Extending the File Mode Using ACLs

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Overview

Through this course:

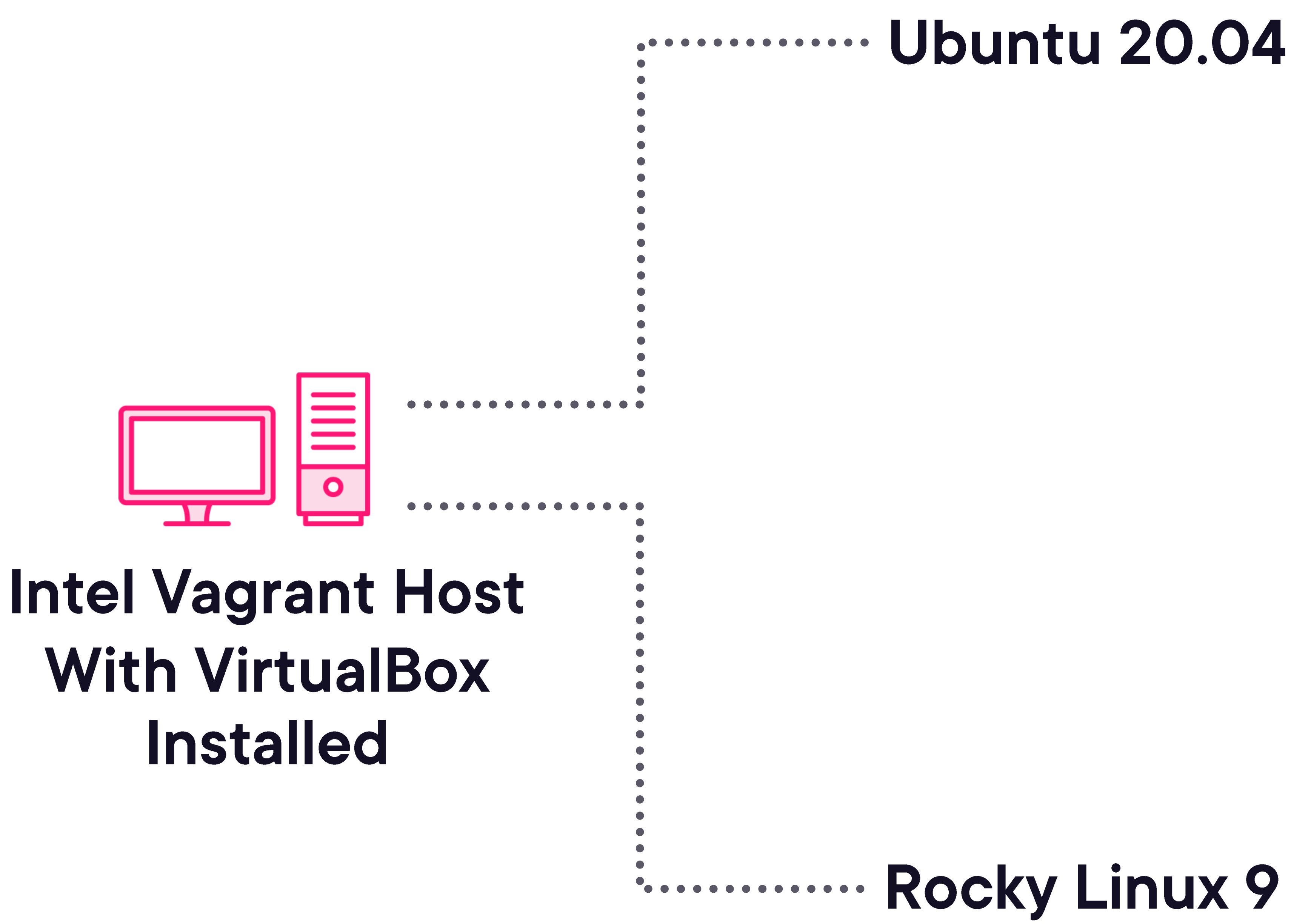
- POSIX ACLS
- Directory Services with OpenLDAP

This Module

- Limitations of the File Mode
- Checking for ACL Support
- ACL Tools
- Setting ACLs
- Setting Default ACLs
- Removing ACLs

Lab Systems

https://github.com/theurbanpenguin/lfcs



LFCS is a certification that is distribution agnostic. Here, you will learn on both Rocky Linux and Ubuntu systems

File Mode / Permissions in Detail

Ubuntu Linux Administration: Essential Commands

Andrew Mallett

File Mode:

- Permissions for one user
- Permissions for one group
- And then, others

```
$ ls -l /etc/hosts
$ stat -c %a /etc/hosts (octal)
$ stat -c %A /etc/hosts (symbolic)
```

Display File Mode

The file mode can be listed with either ls or stat

Demo

Understanding the Limitations of the File Mode:

- Listing the file mode
- SELinux context shows as trailing dot to the file mode

The UNIX file mode was never designed for enterprise file sharing

Allowing for a single user, single group, and everyone else

To work around this, you *can* just keep creating groups to meet new needs in the file system

Even so, this does not cater for when a group requires read access and another group requires read-write access to the same file or directory

ACLs overcome these limitations

POSIX ACLs

Access Control Lists allow for more than one user or group to have the same or similar permissions to a file resource. We can also set default permissions allowing new files or directories to inherit from the parent

Establishing Support for ACLs

yum list acl / apt list acl

grep -i acl /boot/config-$(uname -r)

rpm -qf /bin/getfacl / dpkg -S /bin/getfacl

sudo tune2fs -l /dev/sda1 | grep -i acl (ext4 only)

Demo

Determine Support for POSIX ACLs:

- Kernel Build
- ACL Package
- Mount Options

Viewing ACLs

Where an ACL is set the standard permissions will display a +
drwxrwx---+ 2 root root 4096 Mar 24 13:14 /private/

The command getfacl reads the ACL
$ getfacl /private

Using the command getfattr (from the attr package) you can see the ACL in a rawer state: getfattr -n system.posix_acl_access /private

Demo

View and Set ACLs:

- Using getfacl to view ACLs
- Install and use getfattr to view ACL

```
$ sudo yum install -y httpd
$ sudo setfacl -m d:u:apache:r,d:o:- /var/www/html
$ sudo echo “Hello” | sudo tee /var/www/html/index.html
$ ls -l /var/www/html/index.html
```

Default ACLs

Default ACLs can be applied only to directories. Useful to ensure services can maintain the correct access to files whilst restricting others. Setting the default ACL on the Apache DocumentRoot will not affect existing content. Create your own new index page and the permissions will be correct

```
$ sudo setfacl -m d:u:apache:r,d:o:- /var/www/html
setfacl <rules> <files>
More than one rule can exist, if so, comma separate them
rule 1 d:u:apache:r
rule 2 d:o:-
```

setfacl

Looking at how we set the ACL in more detail:
-m modifies the ACL -- d: in the rule specifies the default ACL
A user rule must include the username
A rule for others does not contain the username. Here we set no permissions for others

Demo

Set Default ACLs:

- Set default ACL on directory
- Using getfacl to view ACLs

Working with ACLs

As root we can create a private directory
$ sudo mkdir -m 700 /private

The user vagrant has no access to this but access is required
$ sudo setfacl -m u:vagrant:rwx /private

We set the default ACL to allow user access to files created within
$ sudo setfacl -m d:u:vagrant:rwx /private

Adding and Removing ACL Entries

We can add additional entries if required

```
$ sudo setfacl -m u:apache:r-x /private
```

If that was an error or we simply needed to remove an entry

```
$ setfacl -x u:apache /private
```

Adding entries for groups is similar to adding users

```
$ setfacl -m g:wheel:r-x /private
```

Demo

Configuring ACLs:

- Adding and removing entries and ACLs
- ACL Mask and Group Mode
- Test and view ACLs

Backing up and Restoring ACLs

To create a backup of an ACL
$ sudo getfacl /private > /tmp/acl.txt

Remove the complete ACL
$ sudo setfacl -b /private

Restore from the / directory. Relative paths are used
$ sudo setfacl --restore=/tmp/acl.txt

Demo

Backup / Copy ACLs:

- getfacl file1 | setfacl --set-file=- file2
- Backup ACL
- Restore ACL

Summary

File Mode

Very restrictive with a single user, group and others

ACL

A list of entries we can manage including setting defaults

Tools

The package acl adds the setfacl and the getfacl tools

Up Next:

Implementing OpenLDAP Directory Services